

## B9W0: African Climate Trend Analysis - Interim Report

### Task 1: Git & Environment Setup

- Created GitHub repository "climate-challenge-week0"
- Set up Google Colab as development environment
- Created README.md and requirements.txt files
- Configured git with user email and username

### Task 2: Data Profiling, Cleaning & EDA

- Uploaded CSV files for 5 African countries (Ethiopia, Kenya, Nigeria, Sudan, Tanzania)
- Replaced -999 values with NaN
- Converted YEAR+DOY to datetime format
- Extracted month for seasonal analysis
- Generated temperature trend visualizations
- Identified Sudan has highest temperature (28.8°C)

### Key Findings:

- Sudan: 28.8°C average (highest)
- Ethiopia: 16.1°C average (coolest)
- Data cleaning successfully handled missing values

### Next Steps:

- Complete cross-country comparison
- Generate extreme heat analysis
- Create vulnerability ranking

### GitHub Repository:

<https://github.com/blen1717/climate-challenge-week0>